INCH-POUND

MIL-C-48397A (AR)
AMENDMENT 6
16 February 2001
SUPERSEDING
AMENDMENT 5
31 January 2000

MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M752A1
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-C-48397A (AR), dated 22 October 1986, and is approved for use by the U.S. Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, Delete "M853", and substitute "M853A1".

PAGE 2

2.1.2, Under “Drawings”, delete:

“Inspection Equipment drawings 9278622 – Envelope drawing for leak testing of ignition cartridges.”

PAGE 4

3.5, Delete in its entirety and substitute the following:

"3.5 Leak tests. Each body assembly, after the primer is loaded and the lacquer cured, and each loaded ignition cartridge, shall satisfactorily pass the leak tests specified in 4.5.4."

3.6.2, Delete in its entirety and substitute the following:

“3.6.2 Velocity and functioning. The cartridge assemblies shall function and shall impart a corrected muzzle velocity as specified in Table II. In addition, the velocity standard deviation of the lot shall be no greater than 3.1 fps.”

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3.7, Delete in its entirety and substitute the following:

"3.7 Propellant charge uniformity. The M9 propellant lots shall achieve a velocity as specified in Table II (see 4.5.7). This test shall be required when a lapse of six months or more occurs between propellant acceptance and LAP initiation."

* Add new paragraph 3.11 as follows:

"3.11 Check test for deterioration of ignition cartridge. If the total elapsed time between original acceptance of the ignition cartridge lot and assembly of that lot into the complete round exceeds two years, or if the ignition cartridges have been subjected to adverse conditions for any period of time since previous tests, the ignition cartridges shall be subjected to, and must satisfactorily comply with the test in 6.6 before the ignition cartridges are assembled into the complete rounds."

PAGE 6

Table I, under "Body Assembly (dwg. 12630551)", delete: "Push-out test …10 …100% …3.2 …4.4.3.1/4.5.1".

Table I, under "Cartridge, ignition, after assembly of cap (dwg. 9294752)", add: "Propellant weight ...25 ...100% ...3.2 ...4.4.3.2.2/4.5.5.2".

PAGE 8

4.4.2.1, Delete Criticals 1 and 2 in their entirety.

Add new Majors 102 and 103 as follows:

"102 Heat shrinkable tubing missing, loose or does not completely cover all flash holes ...100% ...3.2 ...Visual/Manual

103 Pin missing ...100% ...3.2 ...Visual".

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4.4.2.2, Major 102: Delete in its entirety.

Major 103: Delete "Gage" and substitute "Visual".

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4.4.2.3, Delete Majors 101, 102, 103, 105 and 106 in their entirety and substitute the following:

"101 Primer missing or inverted ...100% ...3.2 ... Visual

102 Primer above flush or more than .015 in. below flush ...0.4% ...3.2 ...Gage

103 Lacquer seal missing or inadequate …( ) ...100% ...3.2 ... Gage/4.5.4.1

105 Staking process inadequate ...2/ …( ) ...3.2 ... Gage/4.4.3.1

106 Primer bulged after staking ...100% ...3.2 ... Visual".

Major 104, Delete "lacquer" and substitute "any adhesive or sealant".

Add new Major 107 as follows:

"107 One or more stakes missing or not full ...1/ ...3.2 ...Visual".

Note "(a)", Delete in its entirety and substitute the following:

"1/ Inspect the tool or one body at the start of each day's production and each hour thereafter. A missing or broken tooth shall be cause for rejection of all parts staked since the last successful inspection.

2/ See 4.4.3.1".

PAGE 11

4.4.2.4, Delete drawing number "9549177", and substitute "7549177".

Major 106, Delete in its entirety.

Note (a): Delete in its entirety and substitute the following:

“(a) Twenty-five randomly selected pellets from each shift production. If a Ohaus moisture determination meter is the method of test, the sample shall consist of the quantity of pellets required to equal not less than 10 grams.”

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4.4.2.5, Delete Critical 2 in its entirety.

Add new Critical 5 as follows:

"5 Firing plug above flush with head surface ...100% ...3.2 ...Gage".

Delete Majors 101, 102, 103 and 104 in their entirety.

Major 106, Delete "Gage" and substitute "Visual".

Major 107, Delete "Pellet missing" and substitute "Pellet missing or broken c/".

Add new Major 111 as follows:

“111 Flash tube assembly missing ...100% ...3.2 ...Visual”.

Add new Note c/ as follows:

"c/ Minor chipping of edges acceptable, no chunks or excessive chipping.".

4.4.2.6, Delete 101 in its entirety and substitute the following:

“101 Propellant weight (after loading) (b) ...100% ...3.2 ...4.4.3.2.1/4.5.5.1”.

Delete Major 102 in its entirety.

Add new Major 103 as follows:

"103 Environmental conditions inadequate ...a/ ...3.2 ...Gage".

Add new Note a/ as follows:

"a/: Temperature and humidity shall be checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected.".

Add new Note b/ as follows:

“b/: See 4.4.3.2.1 for defect description. Same sample for both inspections.”

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4.4.2.7, Major 101, Delete examination callout in its entirety, and substitute the following:

"Channel not completely filled with RTV sealant".

Major 104, Delete in its entirety.

Major 105, Delete in its entirety and substitute the following:

"105 True position of outside diameter with major diameter of head thread ...0.40% ...3.2 ...Gage".

Major 106, Delete in its entirety and substitute the following:

“106 Propellant weight ...c/ ...3.2 ...4.4.3.2.2/4.5.5.2”.

Major 107, Delete “4.5.4” and substitute “4.5.4.2”.

Add new Majors 110, 111 as follows:

"110 Environmental conditions inadequate ...b/ ...3.2 ...Gage

111 Presence of excessive sealant ... 0.40% ... 3.2 ... Visual".

Add new Note b/ as follows:

"b/ Temperature and humidity shall be checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected.".

Add new Note c/ as follows:

“c/ See 4.4.3.2.2 for defect description.”.

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4.4.3.1, Delete in its entirety and substitute the following:

"4.4.3.1 Staking process (primer security). Each hour, and on the last units staked before a scheduled staking tool change, five (5) primed bodies shall be inspected before and after staking, using any method that can verify that the primer cups have been pushed down as a result of the staking operation. If none of the primer cups have been pushed down, all units

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loaded since the last successful inspection shall be rejected. If one or more primer cups have been pushed down by any amount, the process may continue unchanged at the discretion of the contractor.".

4.4.3.2, Delete in its entirety and substitute the following:

"4.4.3.2 Weight of Propellant"

4.4.3.2.1, Delete in its entirety and substitute the following:

"4.4.3.2.1 Prior to assembly of cap (see dwg. 9294752). The propellant charge shall be volumetrically dispensed (or tare-weighed) in a container, transferred to the ignition cartridge and then check-weighed 100 percent. The check-weighing shall be as specified in paragraph 4.5.5.1. The volumetric dispensing (or initial weighing) and check-weighing shall be performed by different operators on different scales. Weight limits for weighing shall be established within the applicable requirements, depending upon the accuracy and precision of the weighing system, to assure compliance with the charge weight requirements. Any charge of bulk propellant failing to meet the applicable requirements shall be classed defective using criteria for critical and major defects as defined in paragraph 4.4.3.2.2 and removed from the lot.".

Add paragraph 4.4.3.2.2 as follows:

"4.4.3.2.2 After assembly of cap (See dwg 9294752). Major defect - more than 1 grain below assessed weight. Major defect - More than 1 grain above assessed weight.315 sample ignition cartridge assemblies shall be randomly selected from each lot for this test. If any sample exceeds the assessed weight by more than 10 grains or it two or more samples exceed the assessed weight by 1.10 to 10.0 grains, the lot shall be rejected. If any sample is under the assessed weight by more than 10 grains, or if two or more samples are under the assessed weight by 1.10 to 10.0 grains, the lot shall be rejected. Testing shall be as specified in paragraph 4.5.5.2.".

4.4.3.3, Delete in its entirety.

4.4.3.3.1, Delete in its entirety.

4.4.3.4.1, Delete "0.025" and substitute "0.25".

4.4.3.4.1.1, Delete "0.040" and substitute "0.40".

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4.5.1, Delete in its entirety.

4.5.2, Delete the text in its entirety and substitute the following:

"The moisture content of the black powder pellets shall be determined in accordance with the procedures specified in MIL-P-223. A sample of 25 black powder pellets or a single sample of not less than 10 grams of pellets shall be randomly selected from each production shift. If any pellet or if the 10 gram sample fails to comply with the requirement cited in paragraph 3.3 herein, all pellets in the production shift represented shall be rejected. If loading with propellant has begun, ignition cartridges loaded with the non-conforming black powder pellets shall be rejected.".

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4.5.3.2, Delete sentence 2 in its entirety and substitute the following:

"The firing pin of the test fixture shall be adjusted so that it impacts the firing plug at 0.025 inch off-center.".

PAGE 24

4.5.4, Delete in its entirety and substitute the following:

"4.5.4 Leak tests."

Add paragraph 4.5.4.1 as follows:

"4.5.4.1 Body assembly. After the lacquer has cured, each primed body assembly shall be tested for leakage around the primer. The pressure differential and duration of the test shall be as specified for the leak test of the fully assembled ignition cartridge (see 4.5.4.2). Any body assembly which fails the test shall be classed defective and removed from the lot."

Add paragraph 4.5.4.2 as follows:

"4.5.4.2 Ignition cartridge. The ignition cartridge shall be subjected to a pressure of 3+1 psig for a duration of five (5) seconds minimum. Any ignition cartridge with a leak rate of 1.5 CC per minute or greater shall be rejected and removed from the lot. The test equipment shall be calibrated at the start of each day's operation and prior to production restarts. A restart results from any break in continuous production whether it is due to equipment stoppage/failure or operator breaks. Both accept and reject standards shall be used for each calibration occasion. If the equipment is out of calibration, the ignition cartridges tested since the last successful

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calibration shall be retested after correction of the equipment. Equipment design shall be submitted for approval (see 4.4.4)."

4.5.5, Delete in its entirety and substitute the following:

"4.5.5 Weight of propellant.".

Add new paragraph 4.5.5.1 as follows:

"4.5.5.1 Prior to assembly of cap. The ignition cartridge subassembly shall be weighed before and after the propellant is loaded. The weight of the propellant charge shall be determined by subtracting the tare weight of the ignition cartridge subassembly from the gross weight (after loading). The contractor may elect to perform the 100 percent check-weighing operation with automated equipment and mandatory process controls. In such cases, process control plans in accordance with data item description DI-P-1604 must be submitted to the technical agency (see 6.10) for approval of the automated operation.".

Add new paragraph 4.5.5.2 as follows:

"4.5.5.2 After assembly of cap. The propellant shall be removed from each ignition cartridge sample, placed on the pan of a precision or analytic balance and weighed. Alternatively, each ignition cartridge assembly may be weighed and reweighed after removal of the propellant: The weight of the propellant charge shall be determined by subtracting the weight of the empty ignition cartridge from the loaded (gross) weight. The weight of the propellant in each sample shall be determined to the nearest 0.01 grain.".

4.5.6, Delete second paragraph in its entirety and substitute the following:

"The sample ignition cartridges shall be assembled to 81mm ctgs, M819 or M835A1 having ctg weights as defined in Table II. The complete round, less propelling charges, shall be fired from a M252 81mm mortar at an elevation of forty-five degrees."

4.5.7, Delete third sentence in its entirety and substitute the following:

"Rounds shall be inert loaded to achieve a total weight as specified in Table II."

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Add Table II as follows:

"TABLE II Cartridge Weights

CTG	CTG WEIGHT	MAX VELOCITY (Para. 3.6.2)	MIN VELOCITY (Para. 3.6.2)	UNIFORMITY VELOCITY (Para. 3.7)
M819 M853A1	10.35 +/- 0.02 lbs 8.9 +/- 0.02 lbs	200 fps 220 fps	180 fps 200 fps	190 fps 210 fps

6.1, Delete "M853", and substitute "M853A1".

6.3, Last sentence, delete: "... AMSMC-QAT-I, Dover, NJ 07801-5000" and substitute the following: "AMSTA-AR-QAA-C, Picatinny Arsenal, NJ 07806-5000.".

* 6.6, Delete in its entirety and substitute the following:

"6.6 Check test for deterioration of ignition cartridge"

* 6.6.1, Delete in its entirety and substitute the following:

"6.6.1 Functioning. The requirement shall be as specified in 3.4.2. A sample of 150 each ignition cartridges shall be used. If one or more primers fail to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.3.2".

* 6.6.2, Delete in its entirety and substitute the following:

"6.6.2, Moisture content of black powder pellet. Twenty-five sample black powder pellets shall be randomly selected from the ignition cartridge lot. During disassembly, the ignition cartridge metal parts shall be visually examined for evidence of corrosion or discoloration. The requirement for black powder moisture content is specified in 6.3 and test shall be performed as specified in 4.5.2. The lot shall be rejected if the moisture content fails to comply with the requirement or evidence of corrosion or discoloration exists on metal parts."

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* 6.6.3, Delete in its entirety and substitute the following:

"6.6.3 Nonfunctioning. The requirement shall be as specified in 3.4.1. A sample of 200 each ignition ctgs shall be used. If one or more primers fail to comply with the specified requirement the lot shall be rejected. The test shall be performed as specified in 4.5.3.1."

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